

Turing - Completeness

Recall the Church-Turing Thesis (1936)

Church says: "No computational procedure will be considered an algorithm unless it can be represented by a Turing Machine."

A Standard TM is equal in expressive power to:

- a two-tape TM
- two infinite ends
- allow nondeterminism

Theorem: Every multi-tape TM has an equivalent single-tape TM.

Proof: we convert a multitape TM M to a single-tape TM S .

Say M has k tapes. S simulates the effect of k tapes. By storing their info on a single tape. Use $\#$ as a delimiter. S must also keep track of multiple heads, which it does with a "dot".



If necessary to add a symbol to tape $l < k$, we shift all symbols to the right of that tape by 1.

We can visit any tape by counting the $\#$'s. $\square$

Theorem: A TM T with the added operation 'do not move' (N) is equivalent in power to a standard TM S .

Proof: we are given T and will create S .

Convert each transition as shown:

$$\begin{aligned} \text{Transition in } T: \quad & \bigcirc \xrightarrow{a \rightarrow b, L} \bigcirc \quad \Rightarrow \quad \bigcirc \xrightarrow{a \rightarrow b, L} \bigcirc \\ \text{Transition in } T: \quad & \bigcirc \xrightarrow{a \rightarrow b, R} \bigcirc \quad \Rightarrow \quad \bigcirc \xrightarrow{a \rightarrow b, R} \bigcirc \\ \text{Transition in } T: \quad & \bigcirc \xrightarrow{a \rightarrow b, N} \bigcirc \quad \Rightarrow \quad \bigcirc \xrightarrow{a \rightarrow b, R} \bigcirc \xrightarrow{v \rightarrow v, L \text{ for each } v \in \Gamma} \bigcirc \end{aligned}$$

We are done. $\square$

Turing Test $\neq$ Turing Machines
(sci-fi, AI)

Turing - Complete = "Can simulate a TM."

PDA ✗

FSM ✗

C++ ✓

Redstone ✓

TM without an "L"? ✗
 TM with a finite tape? ✗ } regular!